

The Complete Droste Maker Program

The following is the source code for *drostemaker.py*; because this program relies on the Python-only Pillow library, there is no JavaScript equivalent for this project in this book:

```
from PIL import Image

def makeDroste(baseImage, stopAfter=10):
    # If baseImage is a string of an image filename, load that image:
    if isinstance(baseImage, str):
        baseImage = Image.open(baseImage)

    if stopAfter == 0:
        # BASE CASE
        return baseImage
    # The magenta color has max red/blue/alpha, zero green:
    if baseImage.mode == 'RGBA':
        magentaColor = (255, 0, 255, 255)
    elif baseImage.mode == 'RGB':
        magentaColor = (255, 0, 255)

    # Find the dimensions of the base image and its magenta area:
    baseImageWidth, baseImageHeight = baseImage.size
    magentaLeft = None
    magentaRight = None
    magentaTop = None
    magentaBottom = None

    for x in range(baseImageWidth):
        for y in range(baseImageHeight):
            if baseImage.getpixel((x, y)) == magentaColor:
                if magentaLeft is None or x < magentaLeft:
                    magentaLeft = x
                if magentaRight is None or x > magentaRight:
                    magentaRight = x
                if magentaTop is None or y < magentaTop:
                    magentaTop = y
                if magentaBottom is None or y > magentaBottom:
                    magentaBottom = y
```